Frank Bernal

www.linkedin.com/in/f-bernal | (562) 652-7871 | mendezfrank42@gmail.com

Technical Skills

Software: Python, Bash, Git, GitHub, CI/CD (Bamboo), Ansible, SALT, Linux, SQL, SRE, Grafana, Networking (TCP/IP, DNS, DHCP, LAN/WAN), Confluence, Ignition (Perception/Vision), IPAM, Jira, Matlab/Simulink
Hardware: Autonomous Vehicles, HIL, CAN, Ethernet, DAQ, Altium, DFM, DFMEA, DMM, High Voltage, Harnessing, Intrepid, JTag, LIN, LTSpice, Low Voltage, Oscilloscope, PCB Assembly, Soldering, Vector (CANalyzer)
Manufacturing: 3DX, 5S (LEAN), Machining, Packaging, PFMEA, PLM, Process Optimization, Product Development, Project Management, Rapid Prototyping, SAP, SCADA, Quality Control, NPI, Data Driven, Issue Resolution, Troubleshooting

Professional Experience

Zoox — **Feb 2026 - Present**
Hardware Software Integration - Hardware Systems Engineer

- Engineered a self-service Firmware Flash Station that automates ECU and client device flashing by pulling images directly from AWS S3, removing a manual bottleneck from the production line.
- Developed a Context Framework for an Agentic Test Asset Knowledge Base, enabling automated retrieval and resolution of test asset issues.

Ford — **May 2025 - Feb 2026**
Advanced EV Development - Prototype Development Engineer

- Supported advanced EV prototype development by managing a HIL (LabCar) test environment, triaging platform issues, maintaining tester configurations, and developing automated flashing tools using JTAG and UDS.

Zoox — **Aug 2020 - May 2025**
Manufacturing Operations - Test and Diagnostics Engineer — Jan 2024 - May 2025

- Drove technical investigations to develop a scalable End-of-Line (EOL) test plan and a “headless” ethernet testing framework for EOL and rework activities, enhancing factory testing efficiency and reliability while eliminating the need for supplemental test hardware and reducing overall costs.
- Managed test infrastructure integration during new mass production factory equipment installations, coordinating Factory Acceptance Testing (FAT) at supplier buy-off and final installation stages to ensure seamless deployment, minimal downtime, and compliance with manufacturing standards.
- Spearheaded program-level discussions to improve company diagnostics, focusing on SW/FW configuration management, factory test automation, and diagnostic tooling.
- Managed diagnostic partners to align vehicle build processes (56 days) with a production goal of 3 vehicles per hour (3JPH), reducing testing time by 53%.

Hardware Software Integration - Hardware Systems Engineer — Sep 2022 - Jan 2024

- Spearheaded the end-to-end development of the "Tester PC", a shared hardware/software platform including custom PCBs, harnessing, and peripherals, adopted from prototyping to deployment across the Manufacturing, Service, and Vehicle Integration teams — including use as a HIL test platform.
- Owned Tier 3 troubleshooting and design improvements for the Manufacturing Tester Fleet — production tooling relied on across the manufacturing org — executing preventative maintenance plans with MFG Operations Engineering. (LRUs, FRUs)
- Collaborated with DevOps, IT, and PLM teams to establish test asset workflows, integrate testers and their subsystems into Zoox's lifecycle management, and implement Ansible group configurations, standardizing provisioning and reducing deployment time by 30%.

Manufacturing Operations - Diagnostics Engineer — Dec 2020 - Sep 2022

- Oversaw change management for mechatronics, EE, and advanced hardware systems impacting GA processes.
- Created a queryable issue resolution database used by R&D and service teams for troubleshooting, supporting FMEA activities and diagnostic fault tree initiatives.
- Conducted hands-on analysis and troubleshooting during design, development, and production phases, optimizing manufacturing workflows for next-gen hardware systems.
- Led daily cross-functional efforts to accelerate vehicle build progression, resulting in an 80% decrease in issue tickets and improved assembly reliability.

Education

Oregon State University - *M.S. in Computer Science* **(Expected Spring 2027)**
California State University Long Beach - *B.S. in Electrical Engineering, Power and Control Systems*